

What Is a Data Page in SQL Server?

A **data page** in SQL Server is a fundamental unit of storage. It is a **fixed-size block of 8 KB (8192 bytes)** used to store table rows, index entries, and various internal metadata. SQL Server manages and organizes all database data using these pages, making them essential to understanding how SQL Server handles storage and performance.

Types of Pages in SQL Server

SQL Server uses different page types depending on the kind of data being stored or tracked:

1. **Data Pages**
 - Store actual table row data.
 - Used for **heap tables** and **clustered index leaf-level data**.
2. **Index Pages**
 - Store index entries.
 - Used for **nonclustered indexes** and **upper levels** of clustered indexes.
3. **Text/Image (LOB) Pages**
 - Store large object data types, such as:
 - TEXT, NTEXT, IMAGE
 - VARCHAR(MAX), NVARCHAR(MAX), VARBINARY(MAX)
 - Stored outside the main data page when size exceeds the page limit.
4. **Global Allocation Map (GAM) Pages**
 - Track used or free **extents** in a file.
 - A GAM page covers around **64,000 extents (~4 GB)**.
5. **Shared Global Allocation Map (SGAM) Pages**
 - Track mixed extents that have free space available.
6. **Page Free Space (PFS) Pages**
 - Track free space, page usage, and allocation status.
 - Used to determine where new rows can be inserted.
7. **Index Allocation Map (IAM) Pages**
 - Track extents allocated to a specific table or index within a file.
8. **Boot Page**
 - Contains metadata required to start the database.
 - Exists at page 9 of the primary data file.
9. **Differential Change Map (DCM) Pages**
 - Track extents changed since the last full backup.
 - Used during **differential backups**.
10. **Bulk Change Map (BCM) Pages**
 - Track extents modified by minimally logged operations.
 - Used during **bulk-logged recovery mode**.

Structure of a Data Page

A data page is internally divided into three main components:

1. **Page Header (96 bytes)**
 - Contains metadata like:
 - Page ID
 - Page type
 - Free space count
 - Object ID
 - LSN (log sequence number)
 - Previous/next page pointers (for linked lists)
2. **Data Rows**
 - Actual rows of table data.

- Stored from the **top of the page downward**.

3. Row Offset Table

- A list of pointers to the location of each row.
- Stored from the **bottom of the page upward**.
- Helps SQL Server quickly locate rows.

Row Storage on Data Pages

- SQL Server attempts to store as many rows as possible within a single 8 KB page.
- Maximum size of a **single row = 8060 bytes** (excluding LOB data).
- If a row is larger due to variable-length fields, SQL Server:
 - Stores part of the data on a **separate LOB page**, and
 - Stores a pointer on the main data page.
- Rows can become **forwarded records** in heap tables when updates cause them to no longer fit on the original page.

Page Allocation in SQL Server

Pages are grouped into **extents**, which are units of 8 pages ($8 \times 8 \text{ KB} = 64 \text{ KB}$).

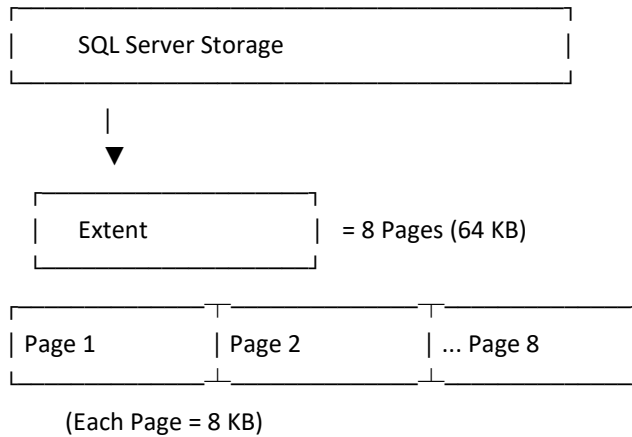
SQL Server uses two types of extents:

1. **Uniform Extents**
 - All 8 pages belong to a **single object** (table or index).
 - Used after an object grows beyond initial allocations.
2. **Mixed Extents**
 - Extent contains pages from **multiple objects**.
 - Typically used when an object is small (e.g., during initial growth).

Additional Important Points

- SQL Server stores pages in **linked lists**, allowing sequential access to data.
- Pages are tracked and organized using internal B-tree structures for indexes.
- Understanding page structure helps diagnose:
 - Fragmentation,
 - Page splits,
 - Storage inefficiencies.
- Tools like **DBCC PAGE**, **DBCC IND**, and **sys.dm_db_database_page_allocations** can inspect page-level data.
- Efficient index design helps reduce page splits and improve I/O performance.
- SQL Server reads data in **8 KB pages**, so reducing page reads (logical/physical) directly improves query performance.

✓ 1. Diagram-Style Explanation (Text Diagram)



Page Layout (Data Page)

Page Header (96 bytes)
• Page ID • Object ID • Free Space Info
• Page Type • LSN • Next/Previous Page Pointers
Data Rows (Stored from top to bottom)
• Row 1
• Row 2
• Row 3 ← Rows stored sequentially
...
Row Offset Table (Bottom-up)
• Offset pointer to Row 1
• Offset pointer to Row 2
• Offset pointer to Row 3

✓ 2. Table Summary of Page Types in SQL Server

Page Type	Purpose	Key Details
Data Page	Stores table row data	Used by heaps & clustered index leaf level
Index Page	Stores index entries	Used in non-clustered and upper-level clustered indexes
Text/Image (LOB) Page	Stores LOB data (VARCHAR(MAX), etc.)	Holds data too large for 8 KB page
GAM Page	Tracks allocated extents	1 GAM covers ~4 GB
SGAM Page	Tracks mixed extents with free space	Helps allocate pages to small objects
PFS Page	Tracks free space & usage	Helps SQL Server find free space for inserts
IAM Page	Tracks extents allocated to an index or table	Links object to physical pages
DCM Page	Tracks extents changed since last full backup	Used for differential backups
BCM Page	Tracks minimal-logged changes	Used in bulk-logged recovery model
Boot Page	Stores database startup metadata	Always at Page 9 of primary MDF

✓ 3. SQL Server Data Page – Quick Cheat Sheet

Page Basics

- Page size = **8 KB (8192 bytes)**.
- Basic unit of storage in SQL Server.
- 8 pages = **1 extent (64 KB)**.

Page Structure

- **Page Header (96 bytes)**
- **Data Rows**
- **Row Offset Table**

Row Rules

- Max row size = **8060 bytes** (excluding LOB).
- Large values → stored in **LOB pages**.
- Heap tables can create **forwarded records** when updates cause rows to move.

Extent Types

- **Uniform Extent** → all 8 pages belong to same table/index.
- **Mixed Extent** → pages can belong to multiple objects.

Important System Pages

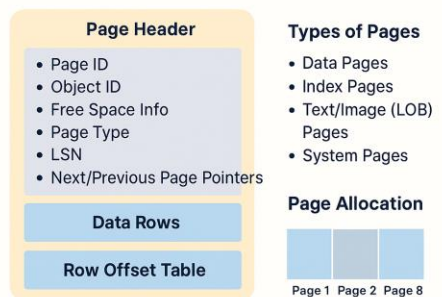
- **GAM** → tracks allocated extents.
- **SGAM** → tracks mixed extents with free pages.
- **PFS** → tracks free space on each page.
- **IAM** → maps extents to objects.
- **DCM** → changed since last full backup.
- **BCM** → minimally logged changes.

Performance Tips

- Proper indexing avoids **page splits**.
- Fill factor affects page fullness.
- Reducing I/O = fewer logical/physical page reads.
- Use DBCC PAGE, DBCC IND for page-level inspection.

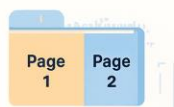
SQL Server Data Page

A data page is a fixed-size block of 8 KB (8192 bytes) used to store table rows, Index entries, and various internal metadata.



Row Storage on Data Pages

- SQL Server tries to fit as many rows as possible on a single 8 KB page
- If a row is too large (e.g. due to VARCHAR(MAX)), it may be stored on a separate page (LOB page)
- Each row has a maximum size of 8060 bytes (excluding LOB data)



Pages are grouped into extents, which are collections of 8 pages (64 KB)

Performance Impact

- Proper indexing avoids page splits

What is a Data Page in SQL Server?

Types of Pages

- **Data Pages** — Store actual table data
- **Index Pages** — Store index entries
- **Text/Image Pages** — Store large object data types like TEXT, NTEXT, IMAGE, VARCHAR(MAX), etc.
- **Global Allocation Map (GAM) Pages** — Track allocated extents
- **Shared Global Allocation Map (SGAM) Pages** — Track mixed extents
- **Page Free Space (PFS) Pages** — Track space usage on pages
- **Index Allocation Map (IAM) Pages** — Track extents allocated to a table or index

Structure of a Data Page

A typical data page contains:

- **Page Header** — Metadata about the page
- **Data Rows** — Actual row data from the table
- **Row Offset Table** — Pointers to the start of each row on page

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